MIDTERM-COMPUTER NETWORKS (ASS-9)

KAMALNAYAK P

24BIT0367

13th Oct 26'

QUESTION NO-2

A) **Develop a Java client–server application to retrieve and display the server's current date and time at the client end.**

SERVER PROGRAM:

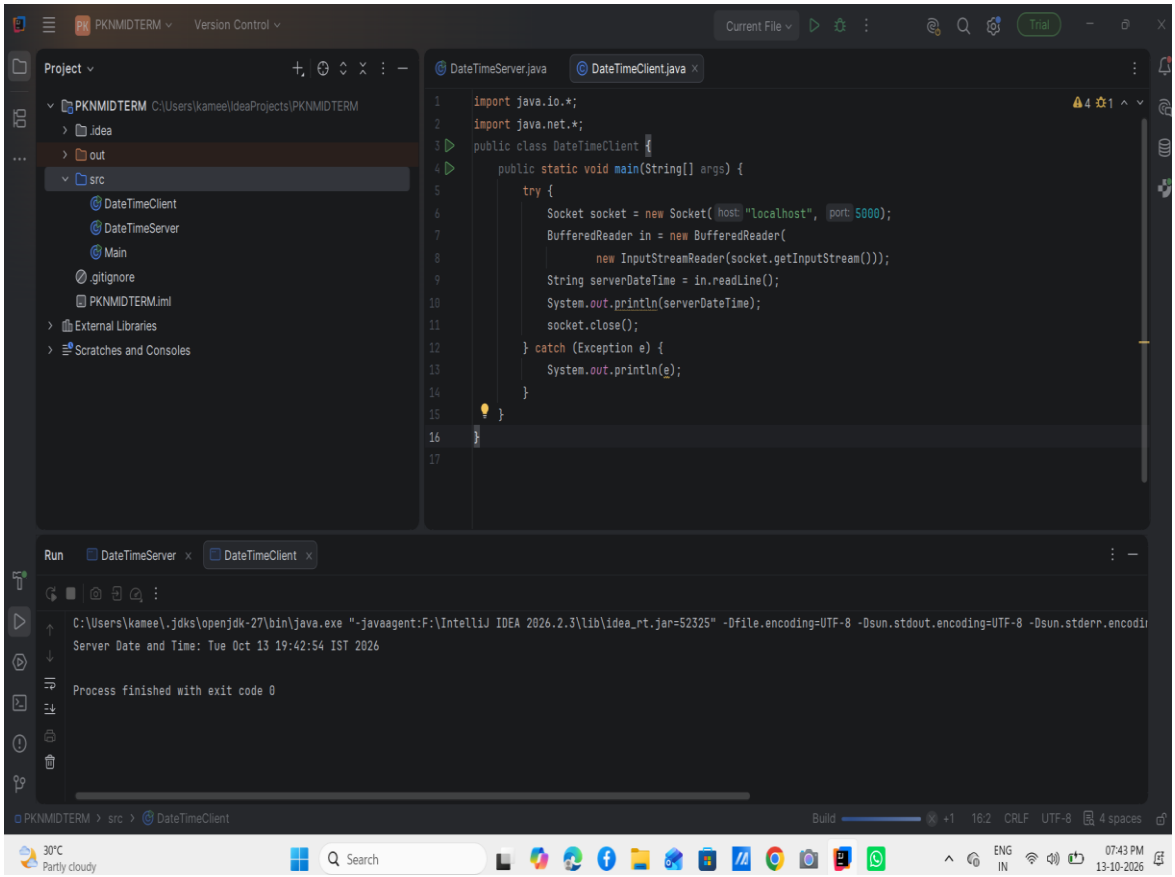
```
import java.io.*;
import java.net.*;
import java.util.Date;
public class DateTimeServer {
    public static void main(String[] args) {
        try {
            ServerSocket serverSocket = new ServerSocket(5000);
            System.out.println("Server is waiting for connection...");
            Socket socket = serverSocket.accept();
            System.out.println("Client connected.");
            PrintWriter out = new PrintWriter(
                    socket.getOutputStream(), true);
            Date currentDate = new Date();
            out.println("Server Date and Time: " + currentDate);
            socket.close();
            serverSocket.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

CLIENT PROGRAM:

```
import java.io.*;
import java.net.*;
public class DateTimeClient {
    public static void main(String[] args) {
        try {
            Socket socket = new Socket("localhost", 5000);
            BufferedReader in = new BufferedReader(
                    new InputStreamReader(socket.getInputStream()));
            String serverDateTime = in.readLine();
            System.out.println(serverDateTime);
            socket.close();
        } catch (Exception e) {
            System.out.println(e);
        }
```

```
        }
}
```

OUTPUT:

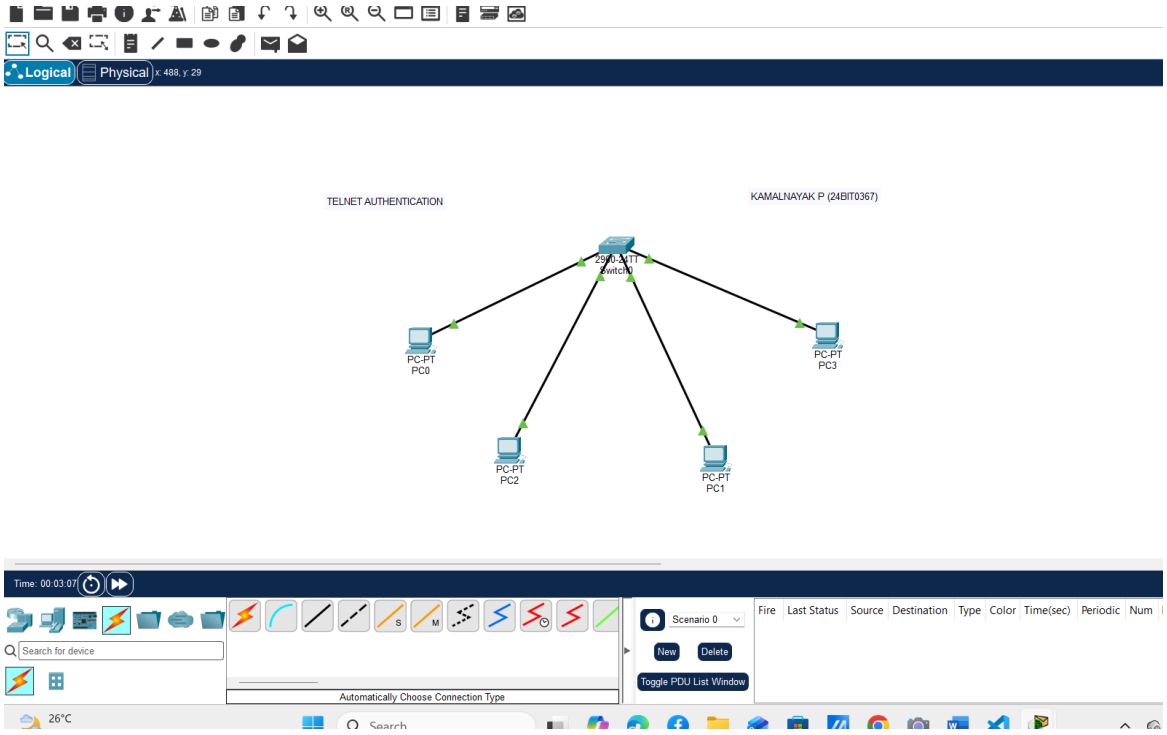

PKNMIDTERM
Version Control
Current File
Trial
Project
PKNMIDTERM C:\Users\kamee\IdeaProjects\PKNMIDTERM
.idea
out
src
DateTimeClient
DateTimeServer
Main
.gitignore
PKNMIDTERM.iml
External Libraries
Scratches and Consoles
DateTimeServer.java
DateTimeClient.java
import java.io.*;
import java.net.*;
public class DateTimeClient {
public static void main(String[] args) {
try {
Socket socket = new Socket(host: "localhost", port: 5000);
BufferedReader in = new BufferedReader(
new InputStreamReader(socket.getInputStream()));
String serverDateTime = in.readLine();
System.out.println(serverDateTime);
socket.close();
} catch (Exception e) {
System.out.println(e);
}
}
}
Run
DateTimeServer
DateTimeClient
C:\Users\kamee\.jdks\openjdk-27\bin\java.exe "-javaagent:F:\IntelliJ IDEA 2026.2.3\lib\idea_rt.jar=52325" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encodi
Server Date and Time: Tue Oct 13 19:42:54 IST 2026
Process finished with exit code 0
PKNMIDTERM › src › DateTimeClient
Build
16:2
CRLF
UTF-8
4 spaces
30°C
Partly cloudy
Search
ENG
IN
07:43 PM
13-10-2026


Run
DateTimeServer
DateTimeClient
C:\Users\kamee\.jdks\openjdk-27\bin\java.exe "-javaagent:F:\IntelliJ IDEA 2026.2.3\lib\idea_rt.jar=60976
Server is waiting for connection...
Client connected.
Process finished with exit code 0

B) TELNET AUTHENTICATION:

STEP-1: OPEN CISCO PACKET TRACER STEP-2: MAKE CONNECTIONS

4PC'S 1 SWITCH



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VTOP Digital Assignment – III (Lab)
Submission date: 18-08-2026
Marks: 10

Subject : BITE305P - Computer Networks Lab
Faculty : Dr. A. Anbarasa Kumar

Lab Digital Assignment Topics

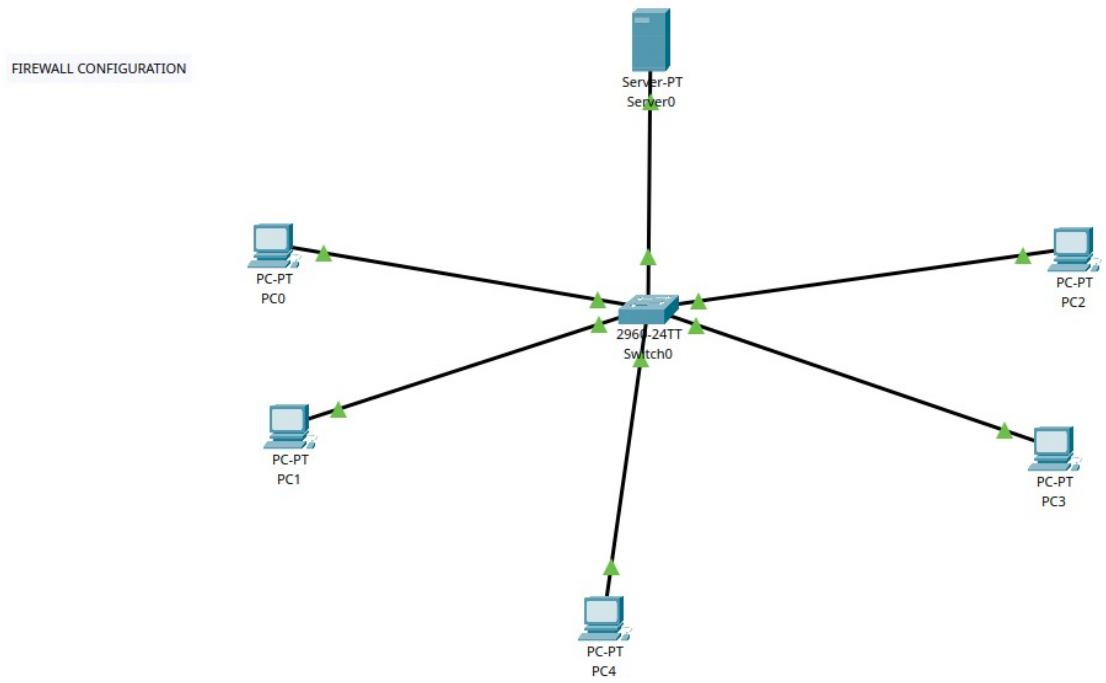
The list of students against the DA topic assigned is given below.

Prepare the document according to the instruction given in the class. Upload your assignment as a **pdf** file in VTOP for assessment before deadline.

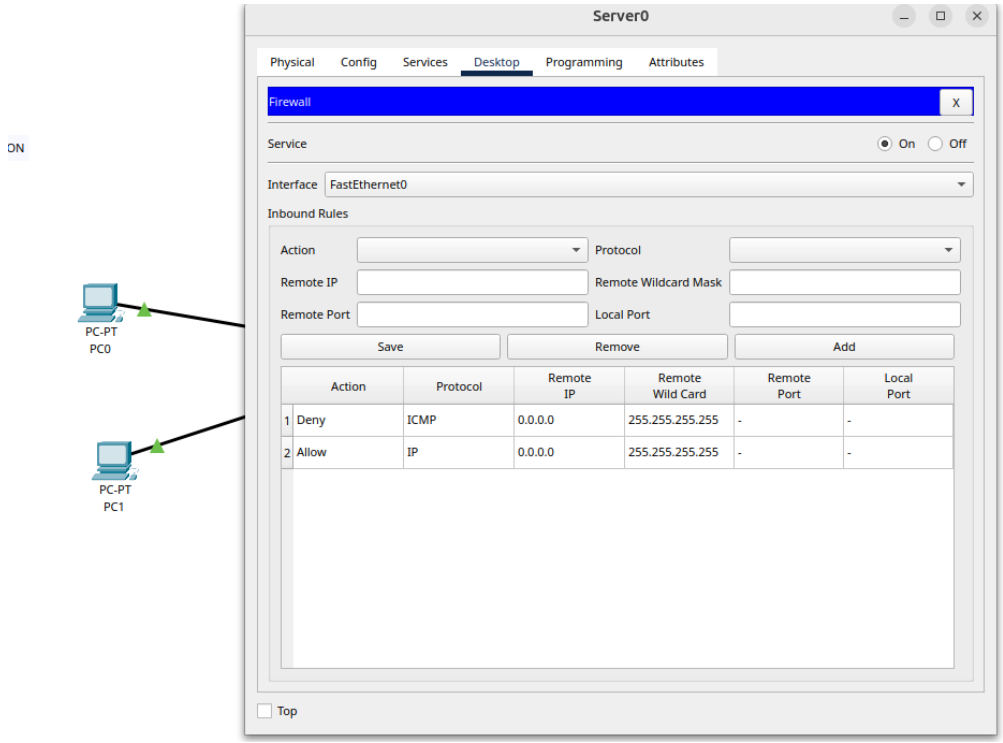
1. To implement and demonstrate the step-by-step procedure for configuring and verifying a firewall using Cisco Packet Tracer. The process opens by Cisco Packet Tracer application and selecting the required network devices. The PCs and server are then configured with appropriate IPv4 addresses and subnet masks based on the given IP addressing table, using a Class C IP address scheme. Firewall are configured on the server to control network traffic, and the network connectivity is verified by testing communication through pinging the IP address of the connected PCs.

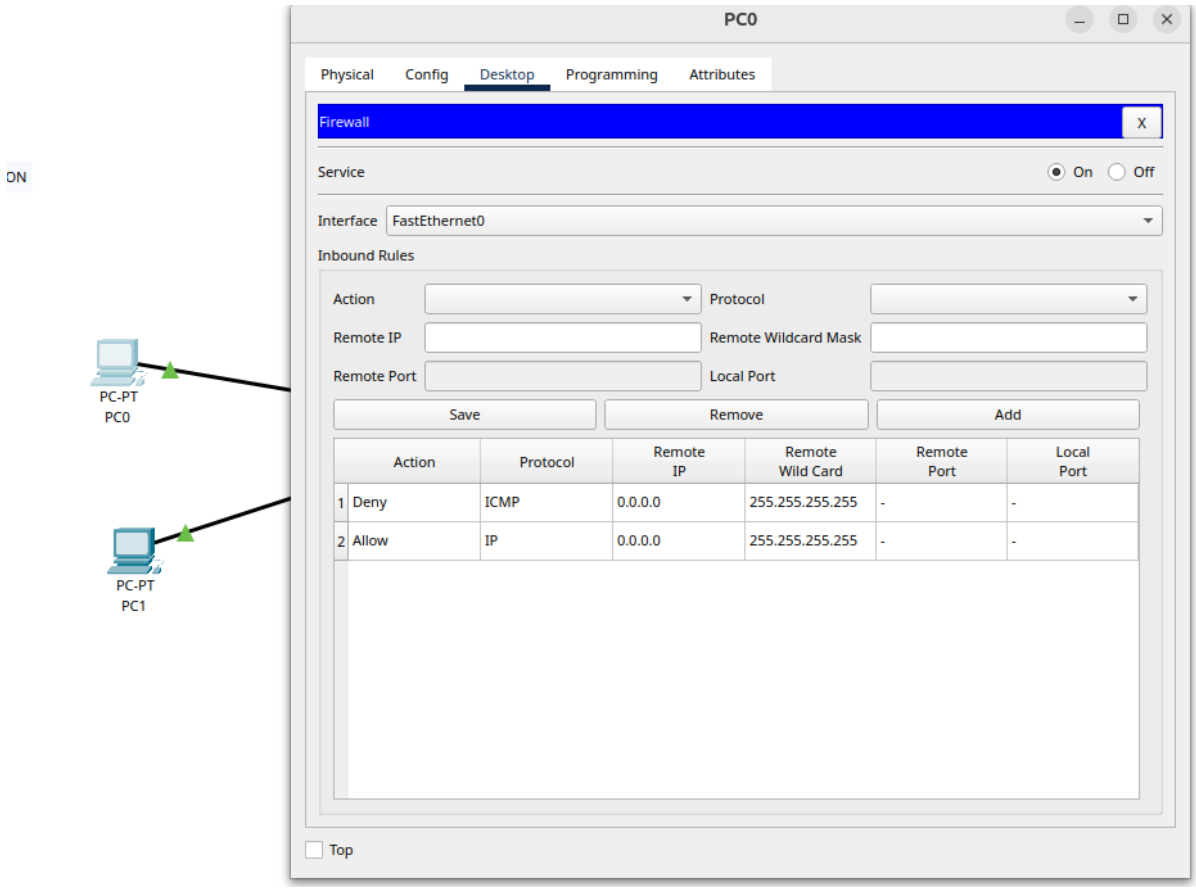
S.No	Device	Model Name	Quantity
1.	PC	PC	05
2.	Server	PT – Server	01
3.	Switch	2960	01

STEP-1: OPEN CISCO PACKET TRACER 2: MAKE CONNECTIONS

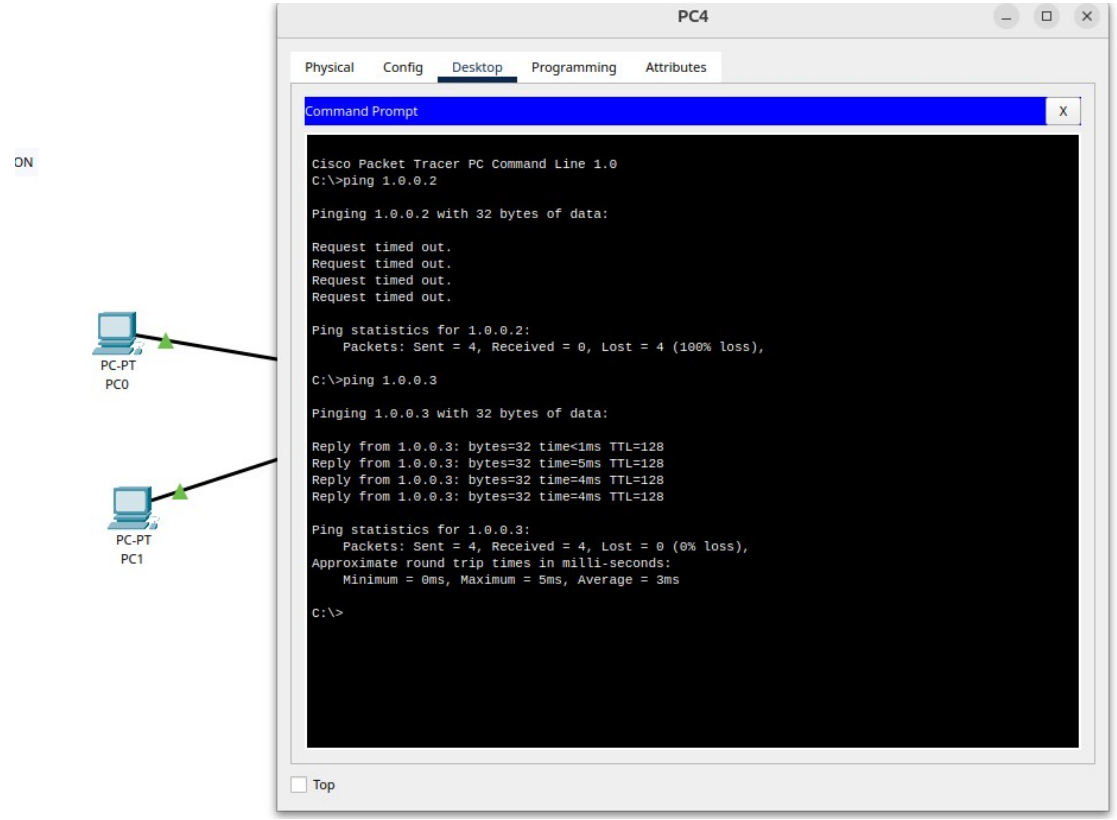


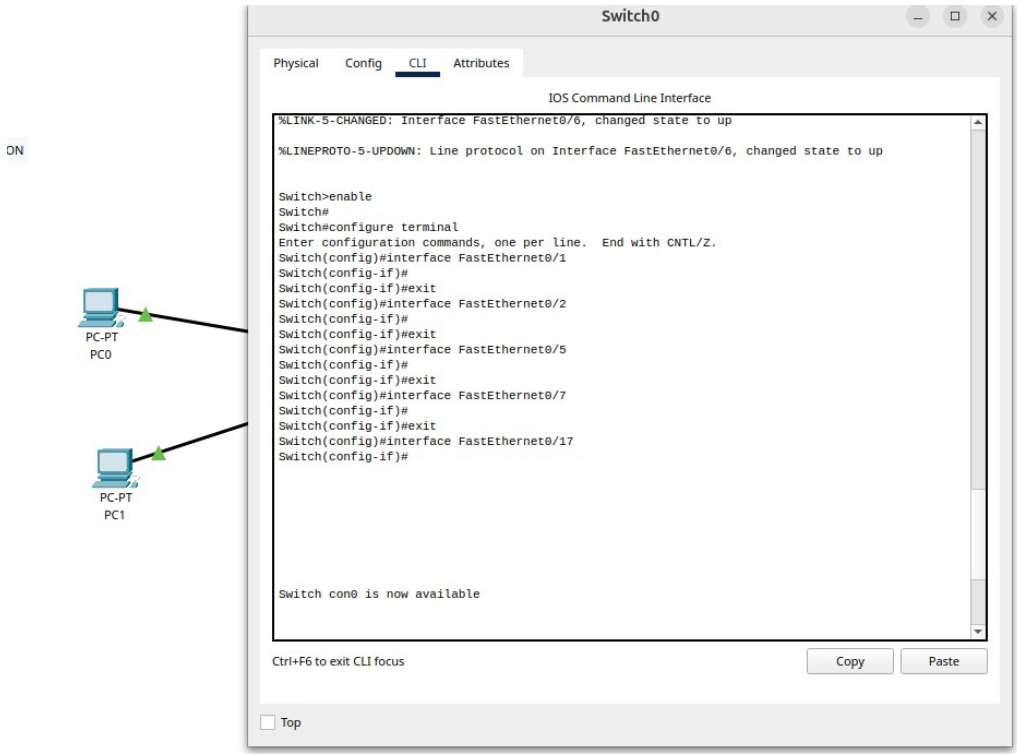
STEP-3: CONFIGURING THE FIREWALL IN SERVER AND THEN PC

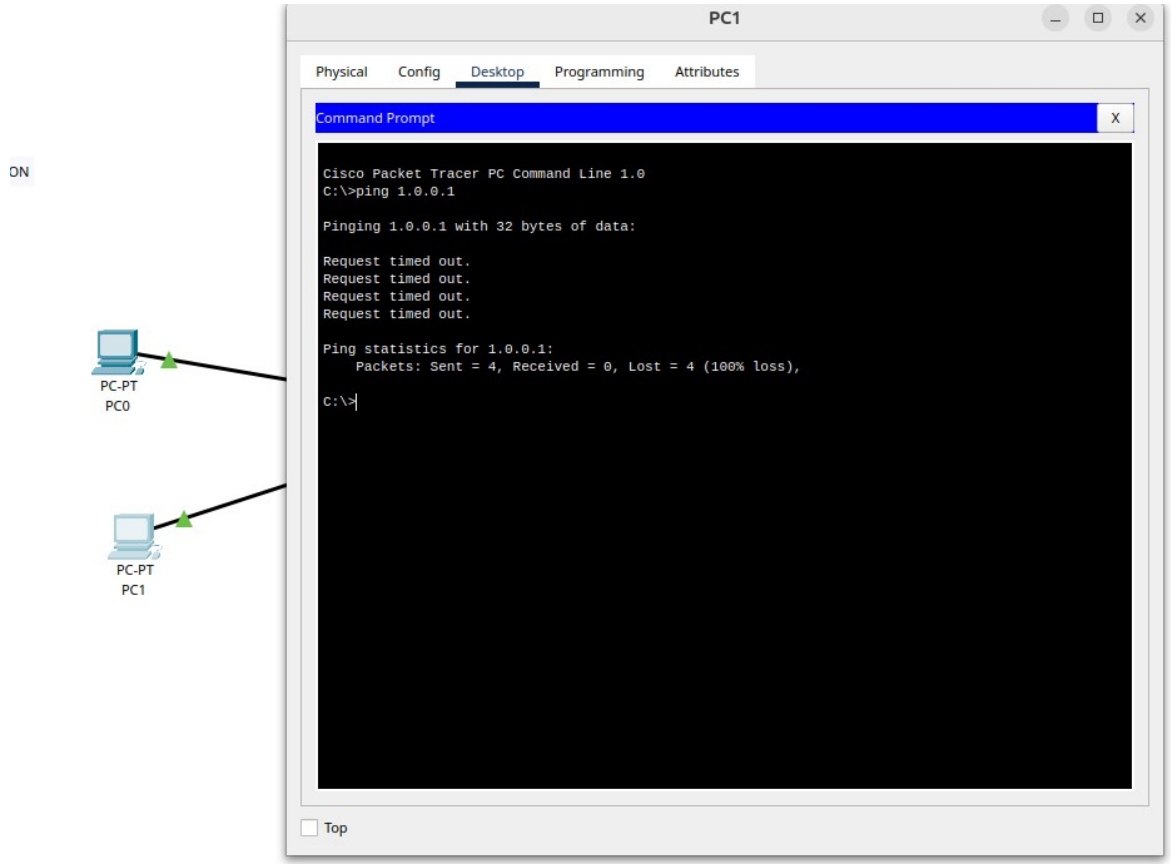




STEP-4: VERIFYING THE NETWORK BY PINGING THE IP ADDRESS OF ANY PC

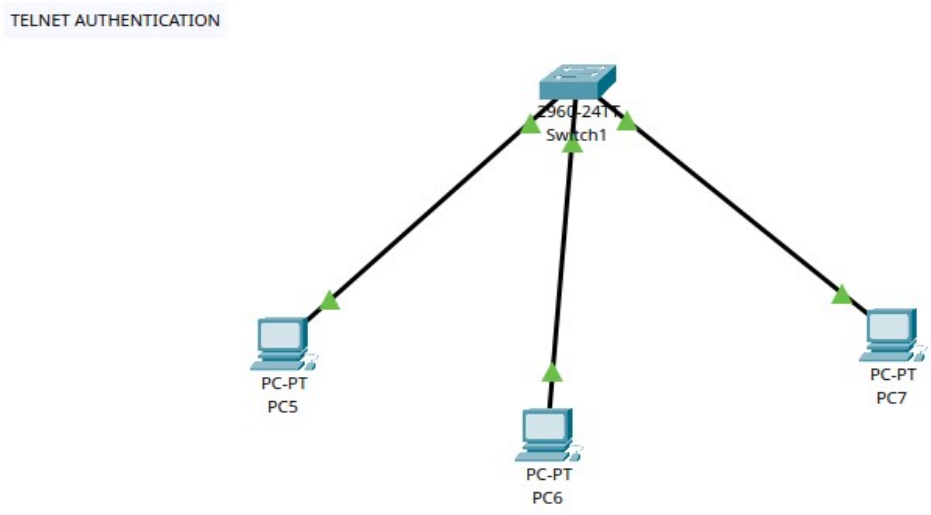




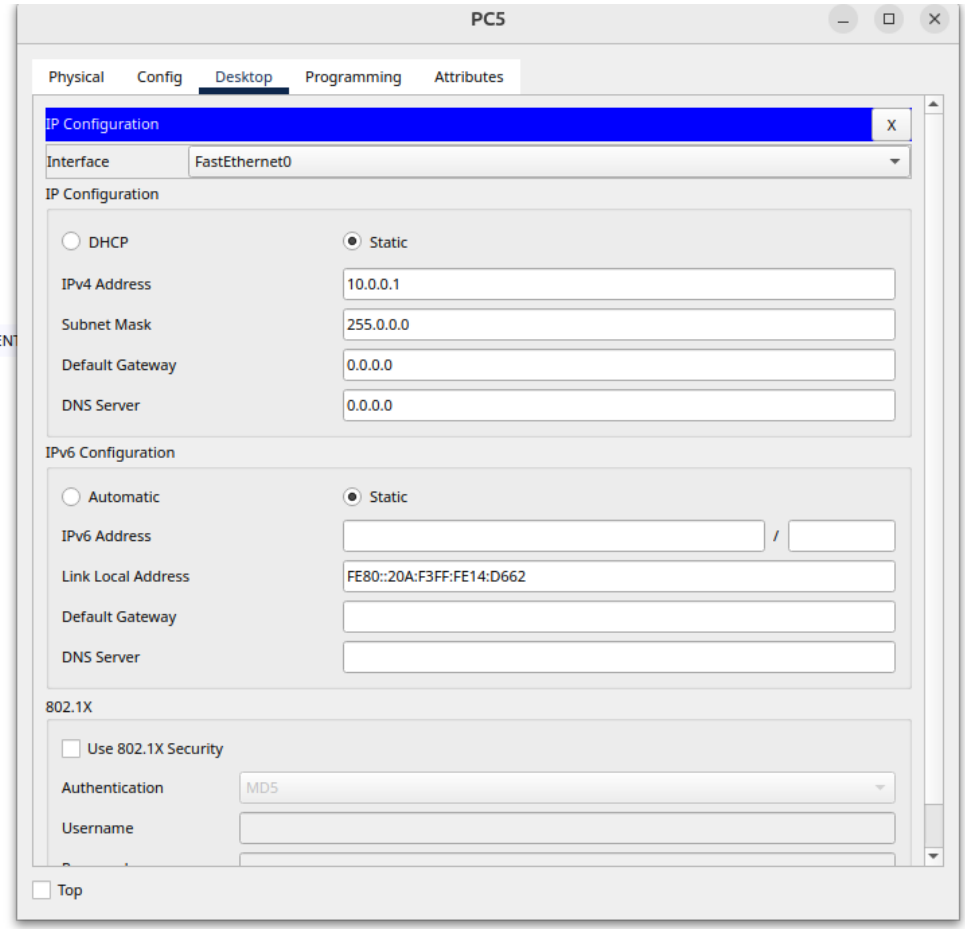


2. Design and implement a network in Cisco Packet Tracer consisting of one Cisco 2960 switch and three PCs. Configure Telnet authentication to enable the secure remote management of the switch. Assign appropriate IP addresses to all devices, configure the switch for Telnet access, and verify remote connectivity from each PC by establishing a successful Telnet session. Demonstrate the complete configuration process and provide the necessary commands used to enable and authenticate Telnet access.

STEP-1: OPEN CISCO PACKET TRACER
STEP-2: MAKE CONNECTIONS

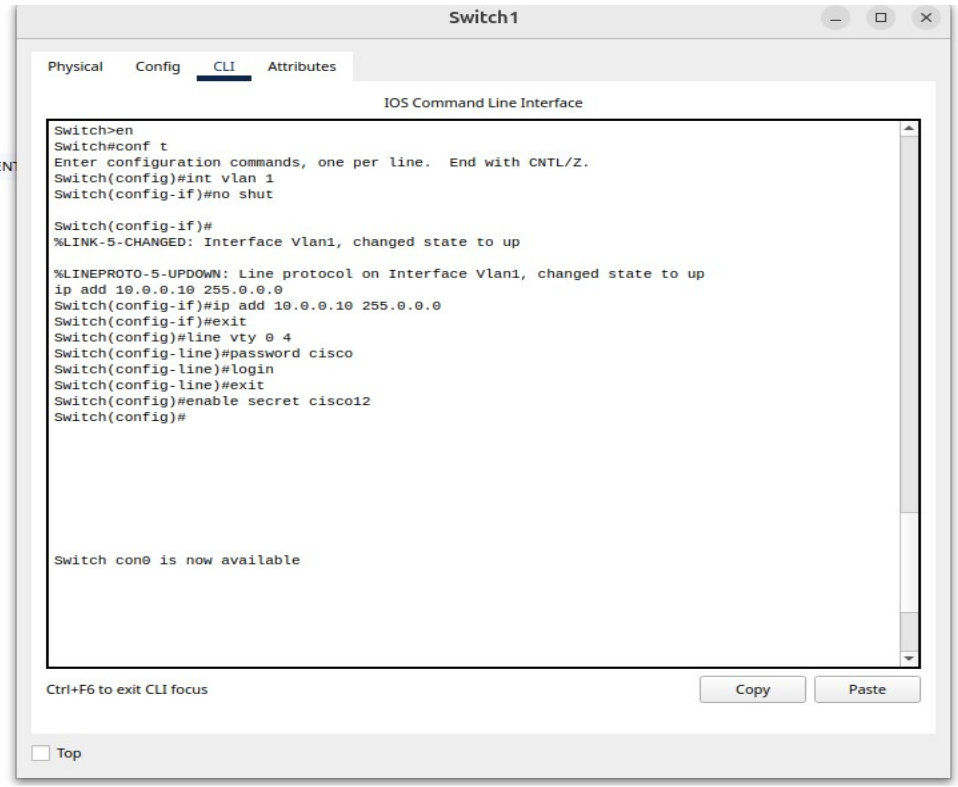


STEP-3: ADD IP ADDRESS TO PC



STEP-4: ENABLE TELNET ON SWITCH

Now hit enter , start writing Commands on CLI.



STEP-5: CHECK TELNET ENABLED OR NOT

